

CD_DA_01

May 1, 2025

1 PySpark analysis flow using a reviews dataset

1.1 Import Libraries and Initialize Spark

```
[3]: from pyspark.sql import SparkSession
import pandas as pd
import numpy as np
# Initialize Spark Session
spark = SparkSession.builder \
    .appName("Amazon Review Analysis") \
    .getOrCreate()
```

1.2 Load The Dataset

```
[5]: # Load the dataset
df = spark.read.csv("amazon_reviews.csv", header=True, inferSchema=True)

# View schema and some rows
df.printSchema()
df.show(5)
```

```
root
 |-- review_id: string (nullable = true)
 |-- product_title: string (nullable = true)
 |-- customer_id: integer (nullable = true)
 |-- review_body: string (nullable = true)
 |-- star_rating: integer (nullable = true)
 |-- review_date: date (nullable = true)
```

```
+-----+-----+-----+-----+-----+
+-----+
|      review_id|  product_title|customer_id|
review_body|star_rating|review_date|
+-----+-----+-----+-----+-----+
+-----+
|24d5a403-1a68-4f7...|    Laptop Stand|      71150|North style woman...|
1| 2025-03-08|
|3b258dc9-98aa-490...|  Fitness Tracker|      27958|Whom be side happ...|
```

```

2| 2025-02-17|
|648b9d29-994d-43c...| Fitness Tracker|      49137|Power head fly pr...|
2| 2024-07-21|
|c54560c5-a776-4d9...| Fitness Tracker|      43921|Reduce only run w...|
4| 2025-02-12|
|cb9ed91e-d288-40e...|Bluetooth Speaker|      69804|Fact nor do deter...|
1| 2024-06-04|
+-----+-----+-----+-----+-----+
----+-----+
only showing top 5 rows

```

1.3 Basic Data Exploration

```

[7]: from pyspark.sql import functions as F

# Count total records
print("Total Records:", df.count())

# Check for missing data in each column
df.select([F.sum(F.col(col).isNull().cast("int")).alias(col) for col in df.
           ↪columns]).show()

# Summary statistics
df.describe().show()

```

```

Total Records: 10000
+-----+-----+-----+-----+-----+
|review_id|product_title|customer_id|review_body|star_rating|review_date|
+-----+-----+-----+-----+-----+
|          0|              0|              0|              0|              0|              0|
+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+
|summary|      review_id| product_title|      customer_id|
review_body|      star_rating|
+-----+-----+-----+-----+
+-----+-----+
| count|          10000|          10000|          10000|
10000|          10000|
| mean|           NULL|           NULL|      55207.0991|
NULL|      3.0092|
| stddev|           NULL|           NULL|25946.418907249783|
NULL|1.417150730028097|
| min|000205b7-2693-44b...|Bluetooth Speaker|      10005|About stock
anyth...|              1|
| max|ffff1bd3-ed38-4b1...|Wireless Earbuds|      99983|useful useful

```

```

ter...|          5|
+-----+-----+-----+-----+-----+
-----+-----+

```

1.4 Data Cleaning

```

[9]: # Drop rows with null reviews
df_clean = df.dropna(subset=["review_body", "star_rating"])
print(df_clean)

```

```

DataFrame[review_id: string, product_title: string, customer_id: int,
review_body: string, star_rating: int, review_date: date]

```

2 Analysis

2.1 Top Reviewed Products

```

[12]: df.groupBy("product_title") \
      .count() \
      .orderBy("count", ascending=False) \
      .show(10)

```

```

+-----+-----+
| product_title|count|
+-----+-----+
| Wireless Earbuds| 2029|
| Smart Watch| 2005|
| Bluetooth Speaker| 1997|
| Fitness Tracker| 1996|
| Laptop Stand| 1973|
+-----+-----+

```

2.2 Average Rating Per Product

```

[14]: df.groupBy("product_title") \
      .avg("star_rating") \
      .orderBy("avg(star_rating)", ascending=False) \
      .show(10)

```

```

+-----+-----+
| product_title| avg(star_rating)|
+-----+-----+
| Fitness Tracker| 3.05811623246493|
| Laptop Stand|3.0440952863659403|
| Smart Watch| 3.000997506234414|
| Wireless Earbuds|2.9990142927550516|

```

```
|Bluetooth Speaker| 2.944416624937406|
+-----+-----+
```

2.3 Most Active Users

```
[16]: df.groupby("customer_id") \
      .count() \
      .orderBy("count", ascending=False) \
      .show(10)
```

```
+-----+-----+
|customer_id|count|
+-----+-----+
|      12436|    3|
|      86231|    3|
|      90777|    3|
|      26478|    3|
|      68828|    3|
|      74826|    3|
|      62401|    3|
|      62534|    3|
|      20317|    3|
|      98665|    3|
+-----+-----+
```

only showing top 10 rows

```
[17]: # Derive Insights

**Most reviewed product is "Wireless Earbuds" with 2029 reviews.**

**Product "Fitness Tracker" has the highest average rating of 3.05.**

**Top 5 users wrote over 500 reviews each.**
```

Cell In[17], line 3

```
**Most reviewed product is "Fitness Tracker" with 224 reviews.**
```

SyntaxError: invalid character "'" (U+201C)